Exercise Sheet 10

Cascades and Temporal Networks

5942UE Network Science

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10.A Cascades and Threshold Rules

Exercise 10.A.1: Threshold Cascade by Hand

Graph (edges A–B, B–C, A–D, C–E, D–F) with thresholds: $A(0.4), B(0.5), C(0.3), D(0.5), E(0.5), F(0.3)$.

Seed at $t = 0$: $\{A, C\}$. Synchronous update.

1. Compute the degree of each non-seed node.
2. Trace which nodes adopt at $t = 1, 2, 3$.
3. Does the cascade reach all nodes?

Exercise 10.A.2: Threshold Heterogeneity

Same graph as 10.A.1. Two experiments:

- **A:** All thresholds raised to 0.6.
- **B:** Only B's threshold raised to 0.6 (others as in 10.A.1).

For seed $\{A, C\}$:

1. Does the cascade reach all nodes in (A)?
2. Does the cascade reach all nodes in (B)?
3. Why does a single threshold change "block" the cascade in (A)?

Exercise 10.A.3: Cascade Simulation

Simulate threshold cascades on `nx.barabasi_albert_graph(100, 3)`.

1. At what seed size does the cascade reliably sweep most of the network?
2. Raise threshold to 0.5 — how does final adoption change?
3. Compare seeding **hubs** (top-k degree) vs **random**.

Exercise 10.A.4: Bridges Differ for Simple vs. Complex Contagion

Dumbbell: two 5-cliques $\{1..5\}$ and $\{6..10\}$ joined by bridge 5–6. Node 1 adopts first.

1. **Simple contagion** ($\beta = 1$): does it spread to clique B?
2. **Complex contagion** ($\theta = 0.4$): does it cross the bridge?
3. What change would let the cascade cross?

10.B Temporal Networks

Exercise 10.B.1: Order-Dependent Reachability

Three nodes A, B, C with two timed edges:

- A–B exists at $t = 1$ only
- B–C exists at $t = 2$ only

1. Can information travel $A \to C$? Trace the path.
2. Can it travel $C \to A$?
3. Reverse the timing (A–B at $t = 2$, B–C at $t = 1$). Does $A \to C$ still work?
4. What does this say about temporal vs. static reachability?

Exercise 10.B.2: Static Aggregation Hides Causal Order

A hospital contact network over 3 days:

Day	Edge
1	Nurse–Patient A
2	Nurse–Patient B
3	Doctor–Nurse

1. Can infection spread from Patient A to Doctor? Trace it.
2. Can it spread from Doctor to Patient A?
3. What might static contact tracing wrongly conclude?

Exercise 10.B.3: Temporal Reachability in Code

Implement BFS over a temporal edge list.

1. From node 0, which nodes are reachable and at what earliest time?
2. From node 3, is node 0 reachable?
3. Add edge $(3, 0, 5)$. How does reachability change?

Exercise 10.B.4: When Timing Kills Spreading

Star with hub H and leaves A, B, C, D. Disease starts at A at $t = 0$.

- **Scenario X:** all edges active at $t = 1$
- **Scenario Y:** H–A at $t = 1$, H–B at $t = 2$, H–C at $t = 3$, H–D at $t = 4$

1. Which nodes get infected in (X)?
2. In (Y)?
3. Now suppose the disease has a **1-step infectious window**. In (Y), does it spread beyond H?
4. What does this illustrate about epidemic containment?

Wrap-Up

- threshold cascades depend on dense local reinforcement, not just connectivity
- bridges are good for simple contagion but bad for complex cascades
- temporal networks make reachability asymmetric and order-dependent
- contact timing relative to infectious windows can decisively contain spread